
DSC 140A - Discussion 05

Problem 1.

Recall that the regularized least squares risk is

$$\tilde{R}(\vec{w}) = \frac{1}{n} \sum_{i=1}^n (\vec{w} \cdot \vec{\phi}(\vec{x}^{(i)}) - y_i)^2 + \lambda \|\vec{w}\|^2$$

Show that

$$\tilde{R}(\vec{w}) = \frac{1}{n} (\vec{w}^T \Phi^T \Phi \vec{w} - 2 \vec{w}^T \Phi^T \vec{y} + \vec{y}^T \vec{y}) + \lambda \vec{w}^T \vec{w},$$

where Φ is the matrix whose i th row is $\vec{\phi}(\vec{x}^{(i)})$, and where $\vec{y} = (y_1, \dots, y_n)^T$.

Problem 2.

In class, we discussed how L1 regularization encourages sparse solutions and can be seen as a method for feature selection. In this problem, we will explore why L1 regularization promotes sparsity from the perspective of gradient descent.

- a) First, write down the partial derivatives of the L1 and L2 regularization terms with respect to a specific weight w_j (you may ignore the case where $w_j = 0$, as the gradient might be undefined there).
- The L1 regularization term is given by:

$$R_1(\vec{w}) = \lambda \sum_{j=1}^d |w_j|$$

- The L2 regularization term is given by:

$$R_2(\vec{w}) = \lambda \sum_{j=1}^d w_j^2$$

- b) Based on these derivatives, which regularizer is more effective at pushing w_j to zero?

Hint: Consider the behavior of the gradients when w_j is already small. For simplicity, assume that the partial derivative $\frac{\partial}{\partial w_j}$ of the Mean Squared Error (MSE) term is zero.